

Requirements Document

Introduction

This document defines the requirements for Shah Sports E-Commerce Platform - a single-vendor sports equipment online store. The system will handle product catalog management, order processing, shipping with multiple carriers (own team + Pathao), promotions/coupons, inventory management, customer email marketing, returns/refunds, and a manual POS system for in-store purchases.

Glossary

- **Admin:** Store administrator with full system access
- **Customer:** End user who purchases products
- **Product:** Sports equipment item available for sale
- **Variation:** Product variant (size, color, etc.)
- **Cart:** Temporary collection of products before checkout
- **Order:** Confirmed purchase transaction
- **POS:** Point of Sale system for in-store transactions
- **Coupon:** Discount code applied at checkout
- **Promotion:** Time-limited discount on products
- **Campaign:** Email marketing initiative targeting customers
- **Shipping_Class:** Category determining shipping rules
- **Weight_Rule:** Shipping cost calculation based on product weight

Requirements

Requirement 1: User Management

User Story: As an admin, I want to manage user accounts, so that I can control access to the system and track customer information.

Acceptance Criteria

1. THE System SHALL support two user roles: Admin and Customer
2. WHEN a user registers, THE System SHALL collect name, email, password, and phone number
3. WHEN a customer registers, THE System SHALL send a verification email
4. THE Admin SHALL be able to view, edit, and deactivate customer accounts
5. WHEN a user logs in, THE System SHALL validate credentials and create a session

6. THE System SHALL support password reset via email

Requirement 2: Category Management

User Story: As an admin, I want to organize products into categories, so that customers can easily browse and find products.

Acceptance Criteria

1. THE System SHALL support hierarchical categories (parent-child relationships)
2. WHEN an admin creates a category, THE System SHALL require name, slug, and optional description
3. THE System SHALL allow categories to have associated images
4. WHEN a category is deleted, THE System SHALL handle associated products appropriately
5. THE System SHALL display categories in navigation menus

Requirement 3: Product Management

User Story: As an admin, I want to manage products with variations, so that customers can purchase sports equipment in different sizes and colors.

Acceptance Criteria

1. WHEN an admin creates a product, THE System SHALL require name, SKU, price, description, and category
2. THE System SHALL support product variations (size, color, weight)
3. WHEN a product has variations, THE System SHALL track inventory per variation
4. THE System SHALL support multiple product images with a primary image designation
5. THE System SHALL associate products with brands and models
6. WHEN a product is marked inactive, THE System SHALL hide it from the storefront
7. THE System SHALL support product SEO metadata (title, description, keywords)

Requirement 4: Brand and Model Management

User Story: As an admin, I want to manage product brands and models, so that customers can filter products by manufacturer.

Acceptance Criteria

1. WHEN an admin creates a brand, THE System SHALL require name and optional logo
2. THE System SHALL associate brands with multiple products
3. WHEN an admin creates a model, THE System SHALL associate it with a brand
4. THE System SHALL allow filtering products by brand and model

Requirement 5: Inventory Management

User Story: As an admin, I want to track product inventory, so that I can manage stock levels and prevent overselling.

Acceptance Criteria

1. THE System SHALL track quantity for each product and variation
2. WHEN inventory reaches a threshold, THE System SHALL notify the admin
3. WHEN an order is placed, THE System SHALL decrement inventory
4. WHEN an order is cancelled, THE System SHALL restore inventory
5. THE System SHALL support manual inventory adjustments with reason logging
6. THE System SHALL display stock status on product pages (in stock, low stock, out of stock)

Requirement 6: Shipping Management

User Story: As an admin, I want to configure shipping options, so that customers can receive products via appropriate delivery methods.

Acceptance Criteria

1. THE System SHALL support two shipping methods: Shah Sports Team (heavy/large items) and Pathao Courier (home delivery)
2. WHEN calculating shipping cost, THE System SHALL consider location, product weight, size, and quantity
3. THE System SHALL support shipping classes to group products with similar shipping requirements
4. THE System SHALL define weight-based shipping cost rules
5. WHEN an order is shipped, THE System SHALL provide tracking information to customers
6. THE System SHALL support free shipping promotions

Requirement 7: Order Management

User Story: As a customer, I want to place orders, so that I can purchase sports equipment online.

Acceptance Criteria

1. WHEN a customer checks out, THE System SHALL create an order with items, shipping address, and payment details
2. THE System SHALL support order statuses: pending, confirmed, processing, shipped, delivered, cancelled
3. WHEN an order status changes, THE System SHALL notify the customer via email
4. THE System SHALL generate unique order numbers
5. THE System SHALL calculate order totals including subtotal, shipping, discounts, and tax
6. THE Admin SHALL be able to view, update, and cancel orders

Requirement 8: Manual POS System

User Story: As an admin, I want to create orders for walk-in customers, so that I can process in-store purchases.

Acceptance Criteria

1. WHEN an admin initiates a POS sale, THE System SHALL allow product selection with quantities
2. THE System SHALL apply available discounts to POS orders
3. WHEN creating a POS order, THE System SHALL collect customer name, email, and optional address
4. THE System SHALL mark POS orders with type "in_store" and payment method "manual"
5. WHEN a POS order is completed, THE System SHALL generate an invoice
6. THE System SHALL send order confirmation and invoice to customer email
7. WHEN a POS order is completed, THE System SHALL update inventory

Requirement 9: Promotions Management

User Story: As an admin, I want to create promotions, so that I can offer discounts to increase sales.

Acceptance Criteria

1. THE System SHALL support promotion types: percentage off, fixed amount off, flash sale, combo offer, free delivery
2. THE System SHALL allow promotions to apply at product level or cart level
3. WHEN multiple promotions exist, THE System SHALL apply only one (no stacking)
4. THE System SHALL support promotion scheduling with start and end dates
5. THE System SHALL allow promotions to target: all products, specific products, specific brands, or specific categories

Requirement 10: Coupon Management

User Story: As a customer, I want to apply coupon codes, so that I can receive discounts on my purchases.

Acceptance Criteria

1. WHEN an admin creates a coupon, THE System SHALL require code, discount type, and value
2. THE System SHALL support coupon formats like "FLASHSALE", "SUMMER2026", "EID2026"
3. THE System SHALL limit coupon usage to once per email address
4. WHEN a coupon is applied, THE System SHALL validate eligibility and calculate discount
5. THE System SHALL support coupon expiration dates
6. THE System SHALL track coupon usage history

Requirement 11: Customer Email Marketing

User Story: As an admin, I want to send marketing emails, so that I can engage customers and promote products.

Acceptance Criteria

1. THE System SHALL support campaign types: promotional, abandoned cart, order updates
2. WHEN creating a campaign, THE System SHALL allow targeting specific customer groups
3. THE System SHALL track email delivery and open rates
4. THE System SHALL support email templates with dynamic content
5. THE System SHALL allow scheduling campaigns for future delivery

Requirement 12: Return and Refund Management

User Story: As a customer, I want to return products, so that I can get refunds for unsatisfactory purchases.

Acceptance Criteria

1. THE System SHALL require product inspection at delivery time
2. THE System SHALL support separate warranty/guarantee policies for specific items (e.g., cardio equipment)
3. WHEN a return is approved, THE System SHALL process refund to original payment method
4. THE System SHALL support partial refunds
5. THE System SHALL track return requests with status updates
6. WHEN a return is completed, THE System SHALL restore inventory

Requirement 13: Reviews and Ratings

User Story: As a customer, I want to review products, so that I can share my experience with other shoppers.

Acceptance Criteria

1. WHEN a customer submits a review, THE System SHALL require rating (1-5 stars) and optional comment
2. THE System SHALL only allow reviews from customers who purchased the product
3. THE Admin SHALL be able to approve, reject, or delete reviews
4. THE System SHALL display average rating on product pages
5. THE System SHALL allow customers to mark reviews as helpful

Requirement 14: Store Policies and CMS

User Story: As an admin, I want to manage store policies and content pages, so that customers can access important information.

Acceptance Criteria

1. THE System SHALL support dynamic store policies (shipping, return, privacy, terms)
2. THE Admin SHALL be able to create and edit CMS pages
3. THE System SHALL support rich text content with images
4. THE System SHALL allow SEO metadata for CMS pages

Requirement 15: Banner and Promotional Content

User Story: As an admin, I want to manage banners, so that I can highlight promotions and featured products.

Acceptance Criteria

1. THE System SHALL support multiple banner positions (homepage, category pages)
2. WHEN creating a banner, THE System SHALL require image, link, and display dates
3. THE System SHALL support banner scheduling and ordering
4. THE Admin SHALL be able to enable/disable banners

Requirement 16: Analytics and Reporting

User Story: As an admin, I want to view analytics, so that I can make data-driven business decisions.

Acceptance Criteria

1. THE System SHALL track sales metrics (revenue, orders, average order value)
2. THE System SHALL report on product performance (views, sales, conversion)
3. THE System SHALL track customer metrics (new vs returning, lifetime value)
4. THE System SHALL provide inventory reports (stock levels, low stock alerts)
5. THE System SHALL support date range filtering for all reports

Requirement 17: Payment Processing

User Story: As a customer, I want to pay securely, so that I can complete my purchase safely.

Acceptance Criteria

1. THE System SHALL integrate SSL payment gateway
2. THE System SHALL support manual payment recording for POS orders
3. WHEN payment is successful, THE System SHALL update order status and send confirmation
4. THE System SHALL store payment transaction records
5. IF payment fails, THEN THE System SHALL display error message and allow retry

Requirement 18: Invoice Generation

User Story: As a customer, I want to receive invoices, so that I can have records of my purchases.

Acceptance Criteria

1. WHEN an order is completed, THE System SHALL generate a PDF invoice
2. THE Invoice SHALL include order details, customer info, items, and totals
3. THE System SHALL send invoice via email to customer
4. THE Admin SHALL be able to regenerate and resend invoices
5. THE System SHALL maintain invoice numbering sequence